Arifcan Demirci

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PROFILE

Mechatronics Engineering graduate and M.Sc. student at Yildiz Technical University with hands-on experience developing robotic software for autonomous mobile robots. **C++** and Python foundation with **object-oriented programming**, multithreading, **Linux** and **Git**, plus **ROS/ROS 2** experience in **navigation, SLAM/mapping, localization**, sensor integration and Gazebo simulation. Built and tested AMR software using LiDAR, IMU, encoders, STM32/Raspberry Pi and UART/I2C communication, and supported real shop-floor commissioning and navigation validation. Developed custom C++ Gazebo plugins and requirements-driven validation workflows, with practical exposure to **Docker/Docker Compose** environments.

EDUCATION

Yildiz Technical University - M.Sc. in Mechatronics Engineering **2026 - Present**

Yildiz Technical University - B.Sc. in Mechatronics Engineering **2022 - 2026**

PEV Science High School **2017 - 2021**

EXPERIENCE

Volunteer Research Assistant **Mar 2025 - Present**

YTU MAGLEV - Istanbul

- Developed **OpenCV**-based camera position estimation and real-time image-processing components; integrated outputs into a **ROS**-based sensing and control architecture.
- Integrated Panasonic HG1200-P analog distance and Hall sensors with dSPACE DS1104 / ControlDesk and Quanser setups; performed data acquisition, measurement validation and control-system integration.
- Built MATLAB/Simulink closed-loop plant/controller models with reference inputs, feedback paths and PID/PD control blocks for response analysis and experimental validation.

Robotics Software Intern **Aug 2025 - Sep 2025**

Patika Robotics - Sakarya

- Built and validated a URDF-based mobile robot model in **Gazebo**, defining link/joint structure, mass-inertia properties, collision geometry and differential-drive plugin parameters.
- Configured and tested **AMCL localization, LiDAR, odometry, TF, global/local costmaps and path planners**; developed Python ROS software for teleoperation and localization-loss detection with safe relocalization.
- Worked with map/odom/base_link coordinate frames and /scan data to validate localization and navigation behavior; reviewed AMCL particle and costmap parameters for stable shop-floor operation.
- Supported field commissioning of a PTK1350 **AMR** at the ASAŞ rolling mill: performed mapping, transferred the map to the control panel, and participated in route-planning and map-based navigation tests on the shop floor.

Robotics Intern **Jul 2024 - Dec 2024**

YTU Computer Engineering - Probabilistic Robotics Research Group, Istanbul

- Worked on ROS-based mobile robot development and integrated LiDAR and camera sensors for environmental perception and system-level testing.
- Tested autonomous navigation and motion-planning algorithms in Gazebo and on physical hardware, comparing simulation with real-system behavior and identifying integration issues.

PROJECTS

Industrial Autonomous Mobile Load-Transport Robot **2026**

ROS 2, Raspberry Pi 4, STM32 NUCLEO-F401RE, RPLIDAR, MPU6050, Encoders, EKF

- Designed a modular architecture separating high-level sensing, localization, navigation and mission logic on Raspberry Pi from low-level motor/encoder control on STM32; implemented bidirectional **UART** at 115200 baud and **I2C**-based IMU acquisition.
- Integrated LiDAR, IMU and encoder measurements; combined odometry and IMU using **EKF sensor fusion** and validated localization outputs in RViz and on the physical prototype, including motion, lift and emergency-stop functions.
- Applied requirements-driven V-Model validation with system/subsystem requirements, test specifications, traceability and pass/fail functional testing.

Magnetic Levitation Based Mobile Module Simulation **2025 - Present**

C++, Gazebo Classic, ModelPlugin, Closed-Loop Control

- Developed a custom **C++ Gazebo ModelPlugin** implementing closed-loop air-gap and orientation control using position, velocity and orientation feedback, PD control, output saturation, AddForce and AddTorque.
- Built a physics-based simulation/validation environment for robot motion, sensor behavior, collision and friction; analyzed system response and iteratively refined the controller and model.

CipperNova - PQFedPredict **2025**

Docker Compose, Federated Learning, JSON Schema, Post-Quantum Security

- Configured a **Docker Compose**-based multi-client simulation environment for distributed federated-learning scenarios and implemented JSON Schema-based validation and model/data-integrity checks.
- Contributed to federated-learning workflows using Flower/FedAvg and post-quantum signed model-transfer concepts, focusing on distributed client behavior, validation and software integration.

2WD Mobile Robot Design and Development **2024**

Embedded Control, Motor Drivers, UART, Bluetooth

- Designed and integrated a low-cost mobile robot platform with DC motor driver electronics and microcontroller-based motion control.
- Implemented **UART**-based Bluetooth communication between the microcontroller and control interface for real-time forward, reverse and turning commands.

TECHNICAL SKILLS

Programming & Software: C++, Python, Embedded C, OOP, multithreading, RAII, smart pointers, resource/memory management, **Git**, **Linux/Ubuntu**

Robotics & Autonomy: ROS, ROS 2, RViz, URDF, **SLAM, AMCL, autonomous navigation**, path planning, TF, odometry, global/local costmaps, EKF, sensor fusion

Simulation & Control: Gazebo Classic, C++ ModelPlugin, MATLAB, Simulink, closed-loop control, PID/PD control, MIL simulation

Communication & Embedded: UART, I2C, TCP/IP, Ethernet, SSH, Modbus fundamentals, STM32, Raspberry Pi, LiDAR, IMU, encoders, cameras

Testing & Tooling: V-Model, requirements analysis, test specifications, requirements-based testing, functional validation, **Docker/Docker Compose**, dSPACE DS1104, ControlDesk, Quanser

ADDITIONAL INFORMATION

Vocational Training: PLC-based Factory Automation Course (48 hours) - Haydarpasa MTAL / Uskudar Innovation Center, Jul 2026. Siemens S7-1200 and TIA Portal fundamentals.

Utility Model: Breadbox System - TURKPATENT No: TR 2023 017890 Y, Inventor, 2026.

REFERENCES

Prof. Dr. Kadir Erkan

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